

MIPT & YSDA · AUTUMN 2026

Deep Generative Models

Lecture 2

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Recap of Previous Lecture

We're given a **finite** number of i.i.d. samples $\{\mathbf{x}_i\}_{i=1}^n \subset \mathbb{R}^m$ drawn from an **unknown** distribution $p_{\text{data}}(\mathbf{x})$.

Objective

Our aim is to estimate a distribution $p_{\text{data}}(\mathbf{x})$ that allows us to:

- **Sampling (generation):** sample $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$.
- **Density evaluation:** evaluate $p_{\text{data}}(\mathbf{x})$ on novel data ("How likely is an object $\mathbf{x}$?").

What is a Divergence?

- $D(\pi \| p) \geq 0$ for all $\pi, p \in \mathcal{P}$
- $D(\pi \| p) = 0$ if and only if $\pi \equiv p$

Divergence Minimization Problem

$$\min_{\theta} D(p_{\text{data}} \| p_{\theta})$$

Recap of Previous Lecture

Forward KL

$$D_{\text{KL}}(p_{\text{data}} \| p_{\theta}) = \int p_{\text{data}}(\mathbf{x}) \log \frac{p_{\text{data}}(\mathbf{x})}{p_{\theta}(\mathbf{x})} d\mathbf{x} \rightarrow \min_{\theta}$$

Reverse KL

$$D_{\text{KL}}(p_{\theta} \| p_{\text{data}}) = \int p_{\theta}(\mathbf{x}) \log \frac{p_{\theta}(\mathbf{x})}{p_{\text{data}}(\mathbf{x})} d\mathbf{x} \rightarrow \min_{\theta}$$

Maximum Likelihood Estimation (MLE)

Let $\{\mathbf{x}_i\}_{i=1}^n$ be i.i.d. observed samples.

$$\theta^* = \arg \max_{\theta} \prod_{i=1}^n p_{\theta}(\mathbf{x}_i) = \arg \max_{\theta} \sum_{i=1}^n \log p_{\theta}(\mathbf{x}_i).$$

Maximum likelihood estimation is equivalent to minimizing a Monte Carlo estimate of the forward KL divergence.

Recap of Previous Lecture

Likelihood as a Product of Conditionals

For $\mathbf{x} = (x_1, \dots, x_m)$, $\mathbf{x}_{1:j} = (x_1, \dots, x_j)$,

$$p_{\theta}(\mathbf{x}) = \prod_{j=1}^m p_{\theta}(x_j | \mathbf{x}_{1:j-1}); \quad \log p_{\theta}(\mathbf{x}) = \sum_{j=1}^m \log p_{\theta}(x_j | \mathbf{x}_{1:j-1})$$

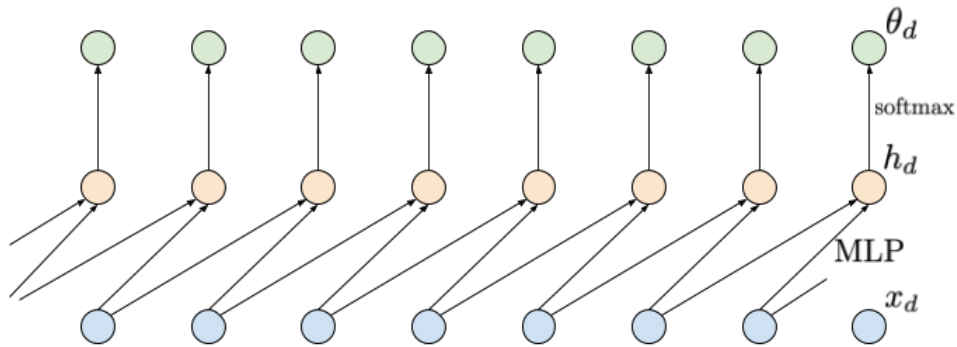
$$\theta^* = \arg \max_{\theta} \sum_{i=1}^n \left[\sum_{j=1}^m \log p_{\theta}(x_{ij} | \mathbf{x}_{i,1:j-1}) \right]$$

Sampling (Ancestral)

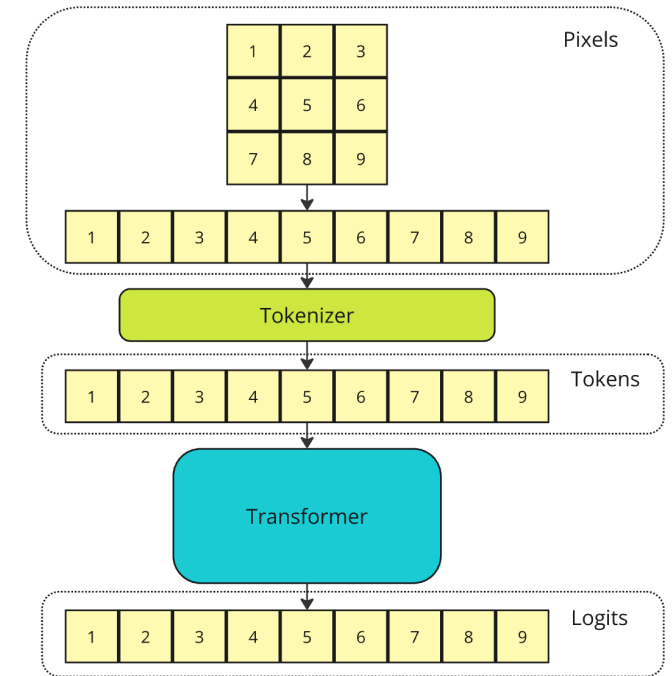
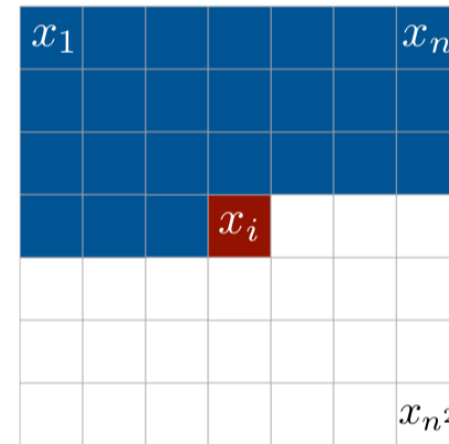
1. Sample $\hat{x}_1 \sim p_{\theta}(x_1)$, $\hat{x}_2 \sim p_{\theta}(x_2 | \hat{x}_1)$, $\dots$, $\hat{x}_m \sim p_{\theta}(x_m | \hat{\mathbf{x}}_{1:m-1})$.
2. Return $\hat{\mathbf{x}} = (\hat{x}_1, \hat{x}_2, \dots, \hat{x}_m)$.

Recap of Previous Lecture

Autoregressive Models: MLP



Autoregressive Models: ImageGPT



Outline

01 Normalizing Flows (NF)

02 NF Examples

- Linear NF

- Gaussian Autoregressive NF

- Coupling Layer (RealNVP)

03 Latent Variable Models (LVM)

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02 NF Examples

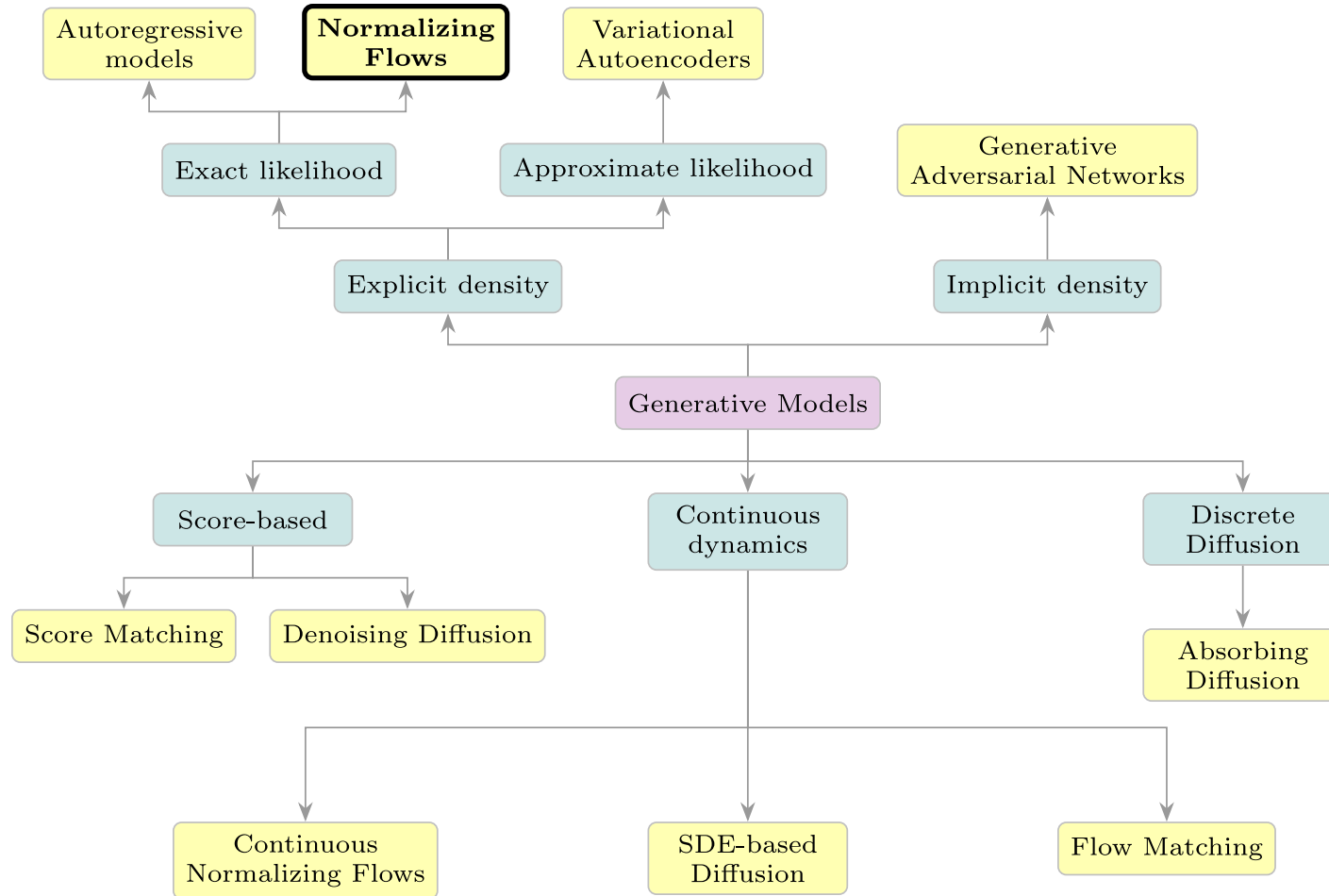
- Linear NF

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Generative Models Taxonomy



Normalizing Flows: Prerequisites

Jacobian Matrix

Let $\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a differentiable function.

$$\mathbf{z} = \mathbf{f}(\mathbf{x})$$

$$\mathbf{J} = \frac{\partial \mathbf{z}}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial z_1}{\partial x_1} & \cdots & \frac{\partial z_1}{\partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial z_m}{\partial x_1} & \cdots & \frac{\partial z_m}{\partial x_m} \end{pmatrix} \in \mathbb{R}^{m \times m}$$

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Change of Variables Theorem (CoV)

Let $\mathbf{x} \in \mathbb{R}^m$ be a random vector with density $p(\mathbf{x})$, and let $\mathbf{f} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a C^1 -diffeomorphism ($\mathbf{f}$ and $\mathbf{f}^{-1}$ are continuously differentiable mappings). If $\mathbf{z} = \mathbf{f}(\mathbf{x})$, then

$$p(\mathbf{x}) = p(\mathbf{z}) |\det(\mathbf{J}_{\mathbf{f}})| = p(\mathbf{z}) \left| \det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) \right|$$

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Jacobian Determinant

Inverse Function Theorem

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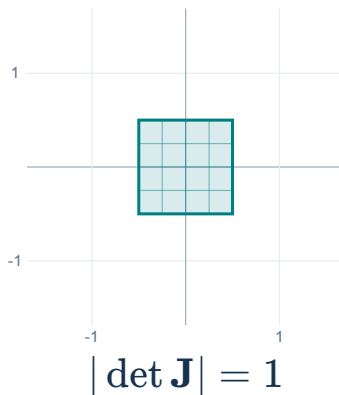
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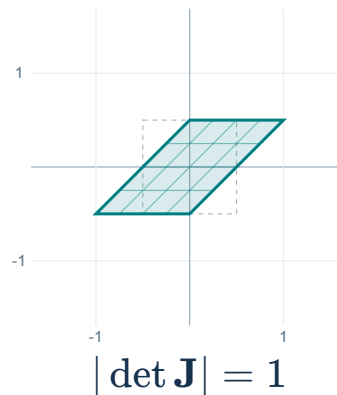
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The determinant of the Jacobian $\mathbf{J} = \frac{\partial \mathbf{f}_{\theta}(\mathbf{x})}{\partial \mathbf{x}}$ quantifies how the volume is changed by the transformation.

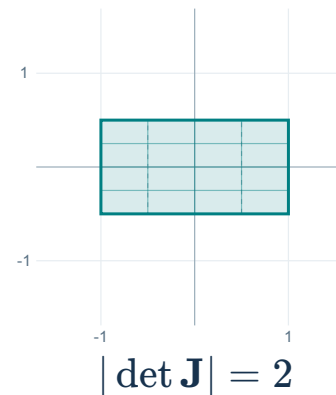
Identity



Shear



Stretch



$$\mathbf{z} = \mathbf{J}\mathbf{x}, \quad \mathbf{J} = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$$

Shear preserves area; stretching by 2 doubles it.

$$p(\mathbf{z}) = p(\mathbf{x}) / |\det \mathbf{J}| \text{ — probability mass is preserved.}$$

Fitting Normalizing Flows

MLE Problem

$$p_{\theta}(\mathbf{x}) = p(\mathbf{z}) \left| \det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) \right| = p(\mathbf{f}_{\theta}(\mathbf{x})) \left| \det \left(\frac{\partial \mathbf{f}_{\theta}(\mathbf{x})}{\partial \mathbf{x}} \right) \right|$$

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Inference

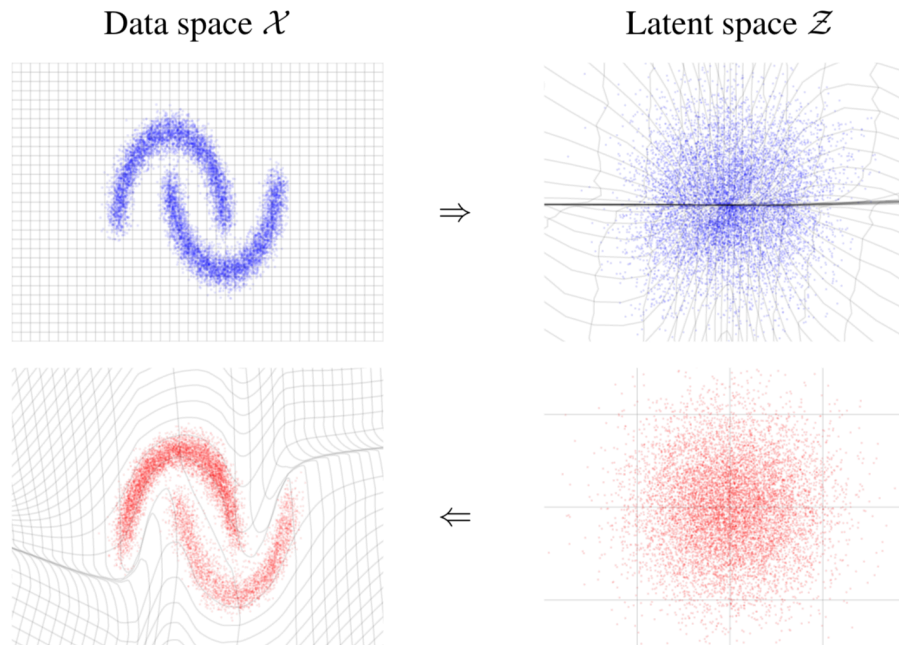
$$x \sim \hat{p}_X$$

$$z = f(x)$$

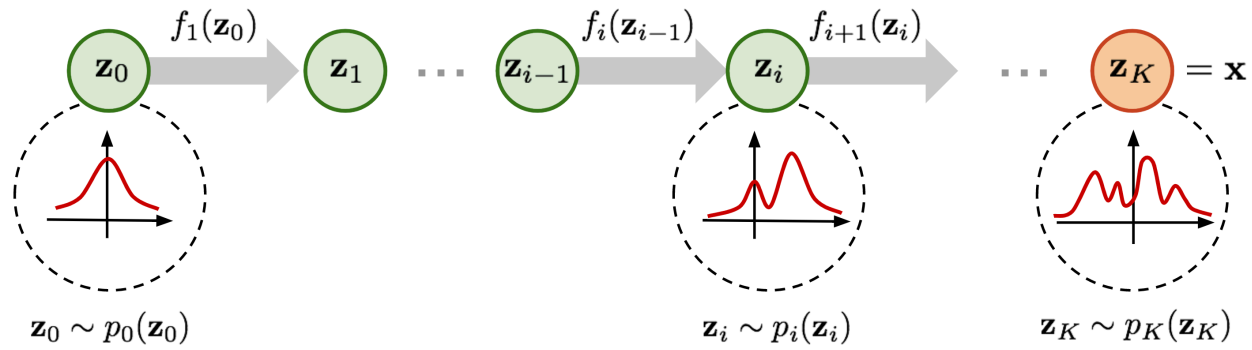
Generation

$$z \sim p_Z$$

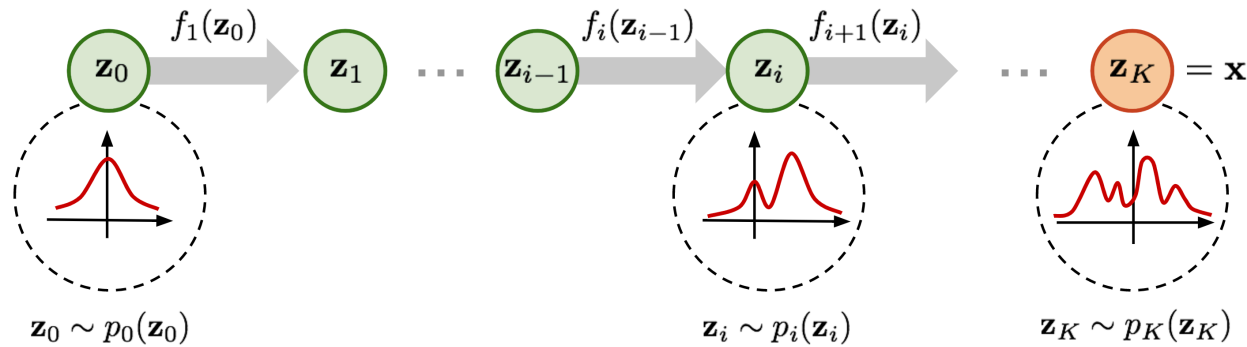
$$x = f^{-1}(z)$$



Composition of Normalizing Flows



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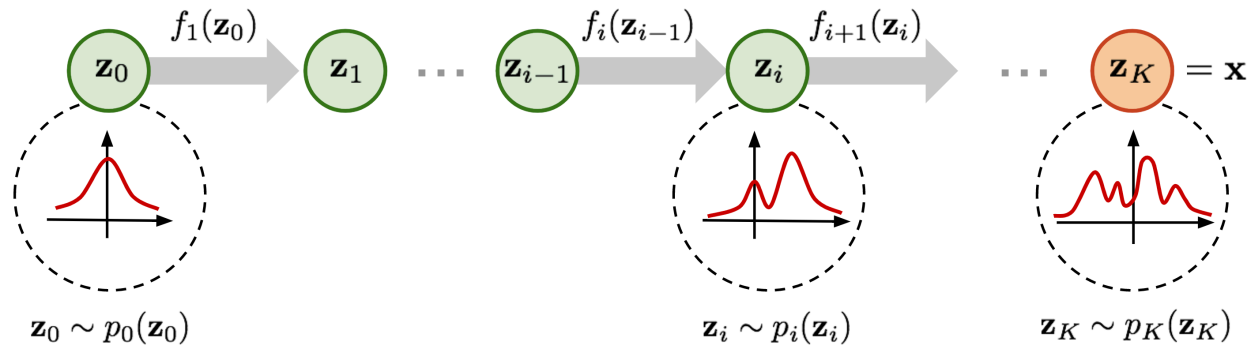


Theorem

If every $\{\mathbf{f}_k\}_{k=1}^K$ satisfies the conditions of the change-of-variables theorem, then the composition $\mathbf{f}(\mathbf{x}) = \mathbf{f}_K \circ \dots \circ \mathbf{f}_1(\mathbf{x})$ also satisfies them.

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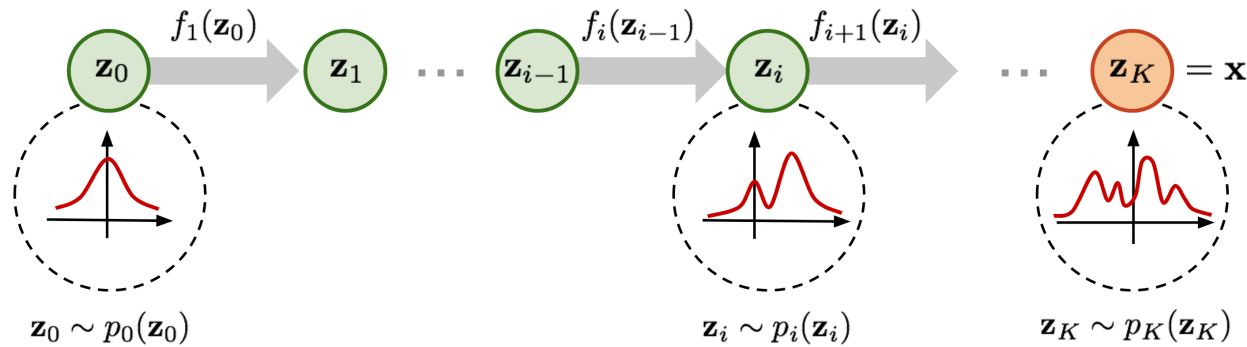


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$$\begin{aligned}
 p_{\theta}(\mathbf{x}) &= p(\mathbf{f}(\mathbf{x})) \left| \det \left(\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}} \right) \right| = p(\mathbf{f}(\mathbf{x})) \left| \det \left(\frac{\partial \mathbf{f}_K}{\partial \mathbf{f}_{K-1}} \cdots \frac{\partial \mathbf{f}_1}{\partial \mathbf{x}} \right) \right| \\
 &= p(\mathbf{f}(\mathbf{x})) \prod_{k=1}^K \left| \det \left(\frac{\partial \mathbf{f}_k}{\partial \mathbf{f}_{k-1}} \right) \right| = p(\mathbf{f}(\mathbf{x})) \prod_{k=1}^K |\det(\mathbf{J}_{\mathbf{f}_k})|
 \end{aligned}$$

Normalizing Flows (NF)

$$\log p_{\theta}(\mathbf{x}) = \log p(\mathbf{f}_{\theta}(\mathbf{x})) + \log |\det(\mathbf{J}_{\mathbf{f}})|$$

Definition

A normalizing flow is a C^1 -diffeomorphism that transforms data $\mathbf{x}$ to noise $\mathbf{z}$.

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- **Normalizing** refers to mapping samples from $p_{\text{data}}(\mathbf{x})$ to a base distribution $p(\mathbf{z})$.
- **Flow** describes the sequence of transformations that maps samples from $p(\mathbf{z})$ to the target, more complex distribution.

$$\mathbf{z} = \mathbf{f}_K \circ \dots \circ \mathbf{f}_1(\mathbf{x}); \quad \mathbf{x} = \mathbf{f}_1^{-1} \circ \dots \circ \mathbf{f}_K^{-1}(\mathbf{z})$$

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Note: Here we consider only **continuous** random variables.

Normalizing Flows

Use a fixed base distribution $p(\mathbf{z})$, e.g. $\mathcal{N}(0, \mathbf{I})$.

Training

1. Sample a minibatch $\mathbf{x}_i \sim p_{\text{data}}(\mathbf{x})$, $i = 1, \dots, n$.
2. Compute $\mathbf{z}_i = \mathbf{f}_{\theta}(\mathbf{x}_i)$ and $\log |\det \mathbf{J}_{\mathbf{f}}(\mathbf{x}_i)|$.
3. Compute loss

$$\mathcal{L} = -\frac{1}{n} \sum_{i=1}^n [\log p(\mathbf{z}_i) + \log |\det \mathbf{J}_{\mathbf{f}}(\mathbf{x}_i)|] .$$

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Sampling

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Requirements

- **Training:** efficient computation of $\log |\det \mathbf{J}_{\mathbf{f}}|$, where $\mathbf{J}_{\mathbf{f}} = \frac{\partial \mathbf{f}_{\theta}(\mathbf{x})}{\partial \mathbf{x}}$.
- **Sampling:** efficient inversion of the transformation $\mathbf{f}_{\theta}$.

What's the computational complexity of evaluating this determinant?

Sampling

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The principal computational challenge is evaluating the Jacobian determinant.

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What is $\det(\mathbf{J})$ in These Cases?

Consider a linear layer $\mathbf{z} = \mathbf{W}\mathbf{x}$, $\mathbf{W} \in \mathbb{R}^{m \times m}$.

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$$\log \left| \det \left(\frac{\partial \mathbf{f}_{\theta}(\mathbf{x})}{\partial \mathbf{x}} \right) \right| = \log \left| \prod_{j=1}^m \frac{\partial f_{j,\theta}(x_j)}{\partial x_j} \right| = \sum_{j=1}^m \log \left| \frac{\partial f_{j,\theta}(x_j)}{\partial x_j} \right|$$

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3. z_j depends only on $\mathbf{x}_{1:j}$ (autoregressive dependency).

Linear Normalizing Flows

$$\mathbf{z} = \mathbf{f}_{\boldsymbol{\theta}}(\mathbf{x}) = \mathbf{W}\mathbf{x}, \quad \mathbf{W} \in \mathbb{R}^{m \times m}, \quad \boldsymbol{\theta} = \mathbf{W}, \quad \mathbf{J}_{\mathbf{f}} = \mathbf{W}$$

Computing $\mathbf{f}_{\boldsymbol{\theta}}^{-1}(\mathbf{z})$ means solving $\mathbf{W}\mathbf{x} = \mathbf{z}$: $O(m^3)$ for a general dense matrix.

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Invertibility

- Diagonal matrix: $O(m)$.
- Triangular matrix: $O(m^2)$.

Why Fix the Determinant Sign?

A continuous path from $\det \mathbf{W} > 0$ to $\det \mathbf{W} < 0$ must cross $\det \mathbf{W} = 0$, where the matrix is **not invertible**.

Therefore, a continuous parameterization $\mathbf{W}(\theta)$, $\theta \in \mathbb{R}^{m^2}$, that guarantees invertibility must keep the determinant sign fixed. It cannot cover **all** invertible matrices.

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Matrix Decompositions

LU Decomposition:

$$\mathbf{W} = \mathbf{P}\mathbf{L}\mathbf{U},$$

- $\mathbf{P}$: permutation matrix.
- $\mathbf{L}$: lower triangular with positive diagonal.
- $\mathbf{U}$: upper triangular with positive diagonal.

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QR Decomposition:

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Linear Normalizing Flows

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- $\mathbf{Q}$: orthogonal matrix.
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Decomposition is performed only at initialization; the decomposed matrices ($\mathbf{P}$, $\mathbf{L}$, $\mathbf{U}$ or $\mathbf{Q}$, $\mathbf{R}$) are optimized during training.

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01 Normalizing Flows (NF)

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Linear NF

Gaussian Autoregressive NF

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03 Latent Variable Models (LVM)

Gaussian Autoregressive Model

Consider the autoregressive model:

$$p_{\theta}(\mathbf{x}) = \prod_{j=1}^m p_{\theta}(x_j | \mathbf{x}_{1:j-1}), \quad p_{\theta}(x_j | \mathbf{x}_{1:j-1}) = \mathcal{N}(\mu_{j,\theta}(\mathbf{x}_{1:j-1}), \sigma_{j,\theta}^2(\mathbf{x}_{1:j-1}))$$

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Generative Direction

$$x_j = \sigma_{j,\theta}(\mathbf{x}_{1:j-1}) \cdot z_j + \mu_{j,\theta}(\mathbf{x}_{1:j-1}), \quad z_j \sim \mathcal{N}(0, 1)$$

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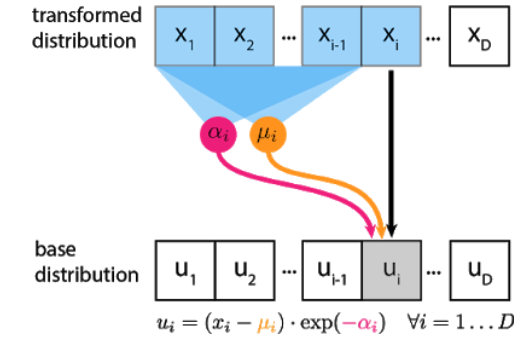
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- The Jacobian matrix of this transformation is triangular.

Gaussian Autoregressive NF

Forward Transformation: $f_\theta(\mathbf{x})$

$$\mathbf{z} = \mathbf{f}_\theta(\mathbf{x})$$

$$z_j = \frac{x_j - \mu_{j,\theta}(\mathbf{x}_{1:j-1})}{\sigma_{j,\theta}(\mathbf{x}_{1:j-1})}$$



Gaussian Autoregressive NF

Forward Transformation: $\mathbf{f}_\theta(\mathbf{x})$

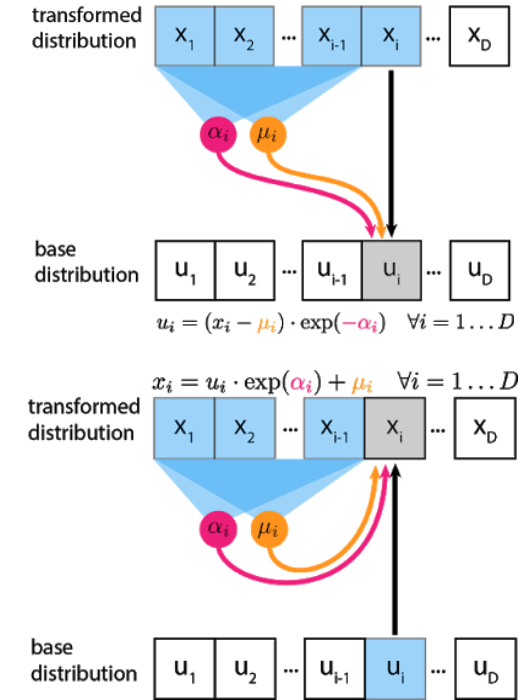
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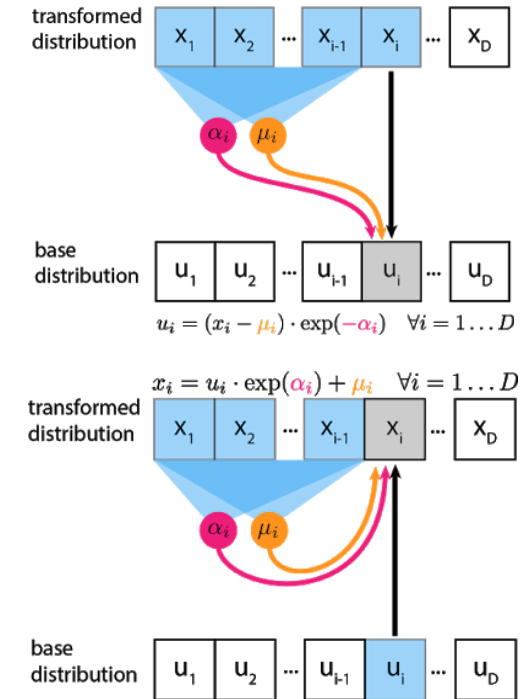
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- Sampling must be done sequentially, but density evaluation can be parallelized.
- The forward KL divergence is a natural objective for training.



Gaussian Autoregressive NF: Two Directions

$\mu_{j,\theta}(\mathbf{x}_{1:j-1}) = \sum_{i<j} x_i, \quad \sigma_{j,\theta} = 1$ A fixed toy Gaussian AR flow

Sampling: sequential

$$x_j = z_j + \sum_{i<j} x_i$$

x_1	x_2	x_3	x_4
-1.000	-0.500	-0.500	-2.500
step 1	step 2	step 3	step 4

Each coordinate needs the previous outputs.

Density evaluation: parallel

$$z_j = x_j - \sum_{i<j} x_i$$

z_1	z_2	z_3	z_4
-1.000	0.500	1.000	-0.500
same step	same step	same step	same step

All prefixes are already available in the observed vector.

The dependency structure is unchanged. What differs is which values are already known.

$\log p_{\theta}(\mathbf{x}) = \log p(\mathbf{z}) + \log |\det \mathbf{J}_{\mathbf{f}}|$; here the Jacobian has unit diagonal, so the log-determinant is zero.

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RealNVP

Split $\mathbf{x}$ and $\mathbf{z}$ into two parts:

$$\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2] = [\mathbf{x}_{1:d}, \mathbf{x}_{d+1:m}]; \quad \mathbf{z} = [\mathbf{z}_1, \mathbf{z}_2] = [\mathbf{z}_{1:d}, \mathbf{z}_{d+1:m}]$$

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Coupling Layer

$$\begin{cases} \mathbf{x}_1 = \mathbf{z}_1 \\ \mathbf{x}_2 = \mathbf{z}_2 \odot \boldsymbol{\sigma}_\theta(\mathbf{z}_1) + \boldsymbol{\mu}_\theta(\mathbf{z}_1) \end{cases}$$

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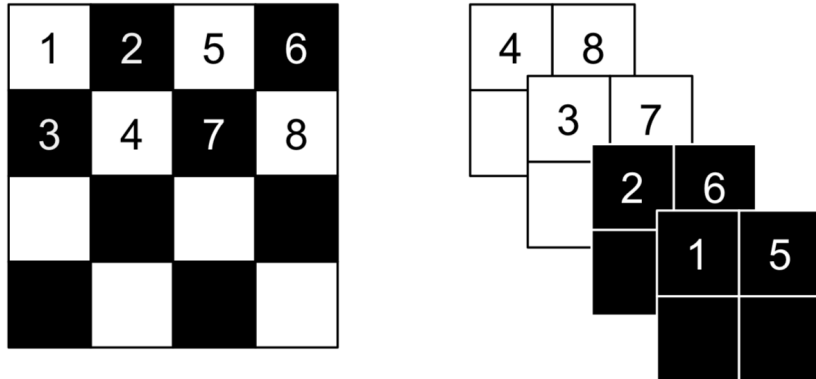
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Image Partitioning



- Checkerboard ordering corresponds to masking.
- Channelwise ordering relies on splitting.

RealNVP

Coupling Layer

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In both training and sampling, only a single forward pass is needed!

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Jacobian

$$\det \left(\frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right) = \det \begin{pmatrix} \mathbf{I}_d & 0_{d \times m-d} \\ \frac{\partial \mathbf{z}_2}{\partial \mathbf{x}_1} & \frac{\partial \mathbf{z}_2}{\partial \mathbf{x}_2} \end{pmatrix}$$

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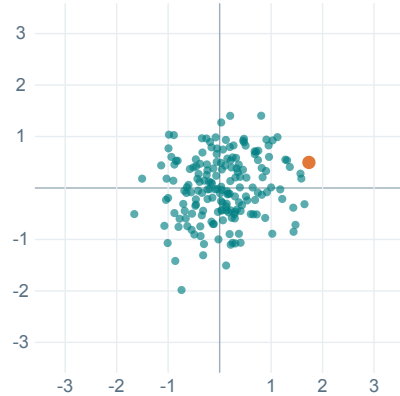
Gaussian AR NF

$$\begin{aligned} \mathbf{x} = \mathbf{f}_{\theta}^{-1}(\mathbf{z}) &\Rightarrow \mathbf{x}_j = \sigma_{j,\theta}(\mathbf{x}_{1:j-1}) \cdot \mathbf{z}_j + \mu_{j,\theta}(\mathbf{x}_{1:j-1}) \\ \mathbf{z} = \mathbf{f}_{\theta}(\mathbf{x}) &\Rightarrow \mathbf{z}_j = (\mathbf{x}_j - \mu_{j,\theta}(\mathbf{x}_{1:j-1})) \cdot \frac{1}{\sigma_{j,\theta}(\mathbf{x}_{1:j-1})}. \end{aligned}$$

How can the RealNVP layer be derived as a special instance of the Gaussian autoregressive NF?

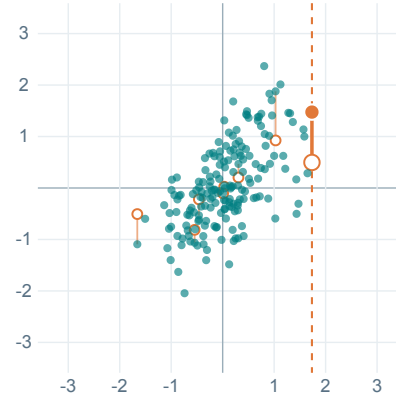
RealNVP: Explore a Coupling Layer

Base noise



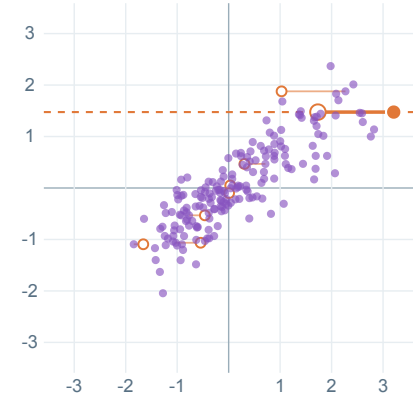
$$\mathbf{z} \sim \mathcal{N}(0, 0.65^2 \mathbf{I})$$

Layer 1: vertical motion



$$y_1 = z_1, \quad y_2 = \sigma(z_1)z_2 + \mu(z_1)$$

Layer 2: horizontal motion



$$x_2 = y_2, \quad x_1 = \sigma(y_2)y_1 + \mu(y_2)$$

$$\sigma(t) = e^{0.4\lambda \tanh(t)}, \quad \mu(t) = 0.8\lambda \tanh(t), \quad \lambda = 1.0 \quad \text{Dashed: fixed coordinate. Hollow: previous position.}$$

$$\frac{\partial \mathbf{y}}{\partial \mathbf{z}} = \begin{pmatrix} 1 & 0 \\ * & \sigma(z_1) \end{pmatrix}$$

$$\frac{\partial \mathbf{x}}{\partial \mathbf{y}} = \begin{pmatrix} \sigma(y_2) & * \\ 0 & 1 \end{pmatrix}$$

Triangular Jacobians.
Positive, explicit scales.

The conditioner stays fixed: undo the shift, divide by the scale. For density evaluation, negate the generative log-determinant.

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Bayesian Framework

Bayes' Theorem

$$p(\boldsymbol{\theta}|\mathbf{x}) = \frac{p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathbf{x})} = \frac{p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{\int p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta})d\boldsymbol{\theta}}$$

- $\mathbf{x}$: observed variables;
- $\boldsymbol{\theta}$: unknown latent variables/parameters;
- $p_{\boldsymbol{\theta}}(\mathbf{x}) = p(\mathbf{x}|\boldsymbol{\theta})$: likelihood;
- $p(\mathbf{x}) = \int p(\mathbf{x}|\boldsymbol{\theta})p(\boldsymbol{\theta})d\boldsymbol{\theta}$: evidence;
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Interpretation

- We begin with unknown variables $\boldsymbol{\theta}$ and a prior belief $p(\boldsymbol{\theta})$.
- Once data $\mathbf{x}$ is observed, the posterior $p(\boldsymbol{\theta}|\mathbf{x})$ incorporates both prior beliefs and evidence from the data.

Bayesian Framework

Consider the case where the unobserved variables θ are model parameters (i.e., θ are random variables).

- $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^n$: observed samples;
- $p(\theta)$: prior distribution.

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Maximum a Posteriori (MAP) Estimation

$$\theta^* = \arg \max_{\theta} p(\theta|\mathbf{X}) = \arg \max_{\theta} (\log p(\mathbf{X}|\theta) + \log p(\theta))$$

Latent Variable Models (LVM)

Maximum Likelihood Estimation (MLE) Problem

$$\boldsymbol{\theta}^* = \arg \max_{\boldsymbol{\theta}} p_{\boldsymbol{\theta}}(\mathbf{X}) = \arg \max_{\boldsymbol{\theta}} \prod_{i=1}^n p_{\boldsymbol{\theta}}(\mathbf{x}_i) = \arg \max_{\boldsymbol{\theta}} \sum_{i=1}^n \log p_{\boldsymbol{\theta}}(\mathbf{x}_i).$$

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The distribution $p_{\boldsymbol{\theta}}(\mathbf{x})$ should be highly complex (just like the true data distribution $p_{\text{data}}(\mathbf{x})$).

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Latent Variable

Introduce a latent variable $\mathbf{z}$ for each observed sample $\mathbf{x}$:

$$p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z}) = p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})p(\mathbf{z}); \quad \log p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z}) = \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) + \log p(\mathbf{z}).$$

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Maximum Likelihood Estimation (MLE) Problem

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The distribution $p_{\boldsymbol{\theta}}(\mathbf{x})$ should be highly complex (just like the true data distribution $p_{\text{data}}(\mathbf{x})$).

Latent Variable

Introduce a latent variable $\mathbf{z}$ for each observed sample $\mathbf{x}$:

$$p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z}) = p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})p(\mathbf{z}); \quad \log p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z}) = \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) + \log p(\mathbf{z}).$$

$$p_{\boldsymbol{\theta}}(\mathbf{x}) = \int p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z}) d\mathbf{z} = \int p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z}.$$

Latent Variable Models (LVM)

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- $p(\mathbf{z})$ is the prior distribution over the latent variable.
- $p_{\theta}(\mathbf{x}|\mathbf{z})$ is the **decoder** distribution.
- Both $p_{\theta}(\mathbf{x}|\mathbf{z})$ and $p(\mathbf{z})$ are usually much simpler than $p_{\theta}(\mathbf{x})$.

Latent Variable Models (LVM)

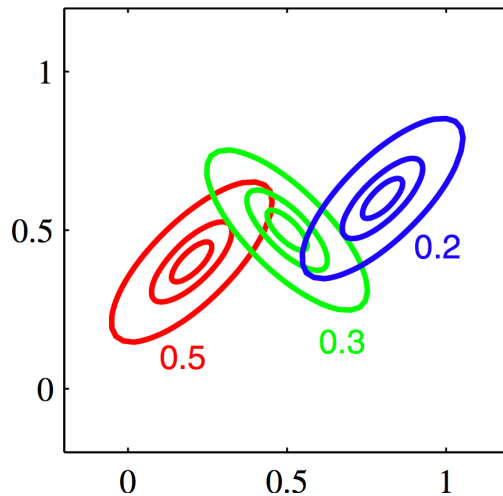
$$\log p_{\theta}(\mathbf{x}) = \log \int p_{\theta}(\mathbf{x}|\mathbf{z})p(\mathbf{z})d\mathbf{z} \rightarrow \max_{\theta}$$

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Examples

Mixture of Gaussians



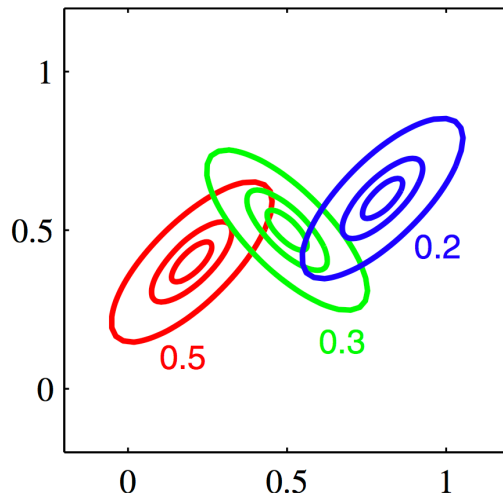
- $p_{\theta}(\mathbf{x}|z) = \mathcal{N}(\boldsymbol{\mu}_z, \boldsymbol{\Sigma}_z)$
- $p(z) = \text{Cat}(\boldsymbol{\pi})$

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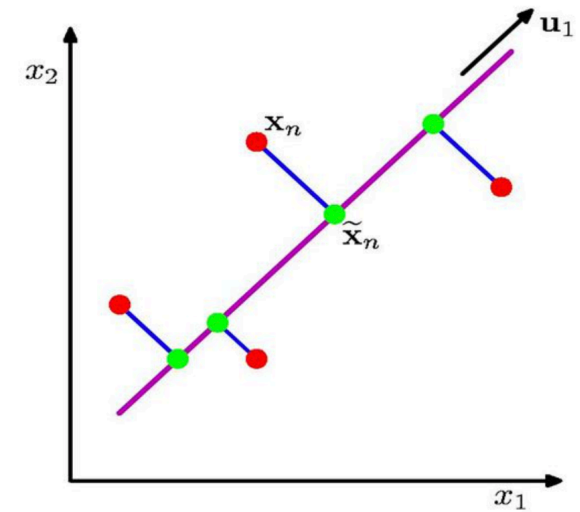
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PCA Model



- $p_{\theta}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{W}\mathbf{z} + \boldsymbol{\mu}, \sigma^2\mathbf{I})$
- $p(\mathbf{z}) = \mathcal{N}(0, \mathbf{I})$

Summary

- The CoV theorem provides a method for computing a random variable's density under an invertible transformation.
- Normalizing flows transform a simple base distribution into a complex one via a sequence of invertible mappings, each with efficient Jacobian determinants.
- Linear NFs capture invertible matrices by using matrix decompositions.
- Gaussian autoregressive NFs are AR models with triangular Jacobians.
- The RealNVP coupling layer provides an efficient normalizing flow (a special case of AR NF), supporting fast density evaluation and sampling.
- LVMs introduce latent representations for observed data, building a complex $p_{\theta}(\mathbf{x})$ from much simpler $p_{\theta}(\mathbf{x}|\mathbf{z})$ and $p(\mathbf{z})$.